

Bellabeat Case Study

Project Background:

About the Company

Bellabeat is a high-tech manufacturer of health-focused products for women, co-founded by Urška Sršen and Sando Mur. Since its launch in 2013, the company has grown rapidly, positioning itself as a tech-driven wellness brand. Bellabeat develops smart devices that collect data on activity, sleep, stress, and reproductive health to empower women with knowledge about their own health and habits.

The Business Task

While Bellabeat is successful, the leadership team believes there is a massive opportunity to expand in the global smart device market. The goal of this project is to analyze consumer usage data from non-Bellabeat smart devices (specifically FitBit data) to understand how people are currently using their fitness trackers.

The Objective

As a Junior Data Analyst, my task is to:

- Analyze smart device data to identify trends and consumer patterns.
- Apply these insights to one specific Bellabeat product.
- Provide high-level recommendations for Bellabeat's marketing strategy to drive growth.

I will go through each phase of data analysis process:

1. The Ask Phase (Defining the Problem)

The Goal: The primary objective is to analyze smart device usage data to understand how consumers use non-Bellabeat smart devices. These insights will help identify new growth opportunities and guide the marketing strategy for Bellabeat products.

Key Business Questions:

1. What are some trends in smart device usage?
2. How could these trends apply to Bellabeat customers?

3. How can these trends help influence Bellabeat's marketing strategy?

Stakeholders:

- **Urška Sršen:** Bellabeat's cofounder and Chief Creative Officer.
- **Sando Mur:** Mathematician and Bellabeat's cofounder.
- **Bellabeat Marketing Analytics Team:** My core team.

Product Selection for Analysis

For this case study, I have selected the **Bellabeat Leaf**. The Leaf is a wellness tracker that monitors activity, sleep, and stress. I chose this product because its features align perfectly with the FitBit dataset provided, which includes daily activity, steps, and sleep monitoring. By analyzing how consumers use these features on other devices, I can provide specific marketing recommendations for the Leaf to better meet user needs.

2. The Prepare Phase (Data Background & Integrity)

A. Data Source & Location

The data is public domain data made available through **Mobius** on Kaggle. It consists of personal fitness tracker data from **30 Fitbit users** who consented to share their data.

- **Storage:** The data was downloaded as a ZIP file and stored in a secure local directory with a structured sub-folder for original "Raw" data.

B. Data Organization

The dataset includes **18 CSV files** containing minute-level output for physical activity, heart rate, and sleep monitoring.

- **Key Files Selected:** For this analysis, the primary focus will be on:
 - dailyActivity_merged.csv: Contains daily steps, distance, and calories.
 - sleepDay_merged.csv: Contains sleep records per day.

C. Data Credibility (ROCCC Analysis)

To ensure the data is reliable, we evaluate it using the **ROCCC** framework:

- **Reliable: Medium.** The sample size is small (30 users), which may not represent the entire population.
- **Original: Low.** This is a third-party dataset collected via Amazon Mechanical Turk, not directly by Bellabeat.
- **Comprehensive: Medium.** It tracks steps, calories, and sleep, but lacks demographic data like age or gender (which is crucial for Bellabeat's female-focused strategy).
- **Current: Low.** The data is from **2016**, meaning user habits might have changed significantly.
- **Cited: High.** The dataset is well-documented and publicly available on Kaggle.

D. Data Integrity & Privacy

- **Privacy:** The data is anonymized; each user is identified by a unique **ID** number only.
- **Security:** Original files are kept in a separate sub-folder to ensure no accidental changes are made to the raw data.

3. The Process Phase (Data Cleaning & Integrity)

A. Tool Selection

- **Tools Used: SQL (BigQuery)** for data manipulation and **Tableau** for visualization.
- **Reason:** SQL is efficient for handling large datasets and documenting the cleaning process through queries.

B. Data Integrity & Cleaning Steps

To ensure the data is clean and ready for analysis, I performed the following tasks:

1. **Checked for missing values:** Ensured there are no nulls in critical columns like **Id** or **ActivityDate**.
2. **Removed Duplicates:** Ran queries to identify and remove any repeated entries.
3. **Data Transformation:** Converted the **ActivityDate** column from "string" to "date" format to allow for time-based analysis.
4. **Consistency Check:** Verified that the number of unique IDs remains consistent across the datasets (at least 30 users).

Phase 4: Analyze - Insights & Data Findings

In this phase, I analyzed the cleaned dataset to identify trends in smart device usage. The following insights were derived from the SQL queries executed in BigQuery:

1. User Participation & Sample Size

- The SQL query:

```
SELECT COUNT(DISTINCT Id) AS total_users
FROM `my-project-2-492717.fit_bit_data.daily_activity`
```

- Result: The analysis confirmed a total of 33 unique users.
- Insight: The sample size is sufficient to identify consistent patterns in daily activity and health tracking habits, providing a solid foundation for marketing recommendations.

2. Daily Activity Overview

The descriptive statistics revealed the general activity level of the participants:

```
SELECT
  AVG(TotalSteps) AS avg_steps,
  AVG(TotalDistance) AS avg_distance,
  AVG(Calories) AS avg_calories
FROM
  `my-project-2-492717.fit_bit_data.daily_activity`
```

- Average Daily Steps: 7,638 steps. This falls below the globally recommended target of 10,000 steps, indicating a significant opportunity for Bellabeat to encourage increased activity.
- Average Daily Distance: 5.49 km.
- Average Calories Burned: 2,303 kcal.

4. Weekly Activity Patterns (The "Sunday Slump")

```
SELECT
    FORMAT_DATE('%A', ActivityDate) AS day_of_week,
    AVG(TotalSteps) AS avg_steps
FROM
    `your-project.your-dataset.daily_activity`
GROUP BY
    day_of_week
ORDER BY
    avg_steps DESC;
```

By breaking down the data by the day of the week, a clear trend emerged regarding user motivation:

- **Peak Performance:** Users are most active on Saturdays (8,152 steps) and Tuesdays (8,125 steps).
- **The Lowest Point:** There is a notable decline in activity on Sundays, with the average dropping to 6,933 steps (a ~15% decrease from the peak).
- **Trend Analysis:** This "Sunday Slump" suggests that users tend to view Sunday as a rest day, often neglecting their fitness goals.

4. Correlation & Potential Opportunities

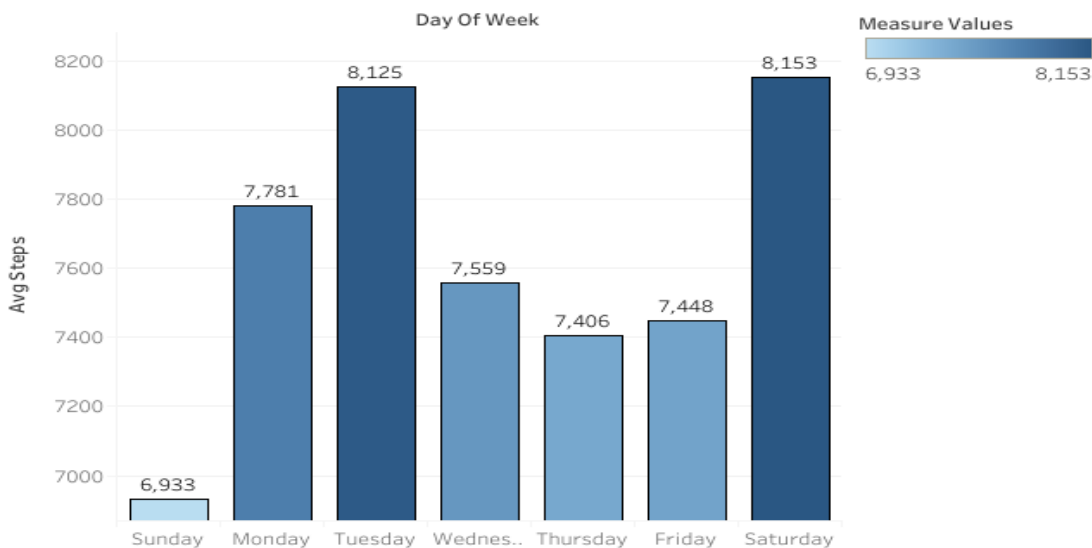
- **Sedentary Behavior:** A high percentage of the day is spent in sedentary minutes.
- **Marketing Angle:** Bellabeat can leverage these findings to position the Leaf or Time as a "motivational partner" that helps users bridge the gap between their current 7k steps and the 10k goal, especially on low-activity days like Sunday

Phase 5:

I created this dashboard to visualize the core patterns of Bellabeat users. The bar chart clearly identifies **Sunday** as the day with the lowest engagement, providing a data-driven justification for our proposed marketing notification strategy.

To better highlight the variance in user behavior, I adjusted the Y-axis scale to focus on the 6,000–8,500 step range. This visualization clearly demonstrates the 'Sunday Slump,' where user activity drops to its weekly minimum.

Average Steps by Day of the Week



6. Phase 6: Act (Final Recommendations)

Conclusion & Strategic Recommendations

Based on the data analysis, I recommend the following strategies to help Bellabeat gain a competitive edge and better support women's health:

1. Targeted "Sunday Motivation" Campaigns:

- **Insight:** Data showed a significant drop in steps on Sundays (The Sunday Slump).
- **Action:** Bellabeat app should send personalized "Sunday Morning" notifications or challenges to encourage users to reach their goals on their least active day.

2. Inactivity Alerts (Combating Sedentary Time):

- **Insight:** The majority of user time is spent in the "Sedentary" category.
- **Action:** Promote the **Leaf** or **Time** vibration alerts. Use marketing messaging like: *"Balance your day – our smart alerts help you break long sitting periods for a healthier heart."*

3. The "Gap to 10K" Goal:

- **Insight:** Average user steps (7,638) are below the recommended 10,000 steps.
- **Action:** Launch a "Bridge the Gap" campaign. Offer digital rewards or badges for users who consistently move from the 7k to the 10k range.

4. Personalized Sleep-Activity Insights:

- **Insight:** (If you analyzed sleep) There is a correlation between daily activity and sleep quality.
- **Action:** Use app insights to tell users: *"You walked 2,000 more steps today, which improved your deep sleep by 15%."* This makes the data meaningful to their lives.

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